Predicting Good Solar Hours for Energy Planning from Weather and Location Data

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Changes since Milestone 1

While preparing Milestone 2, found and corrected the following:

1. **Altitude unit.** Altitude was converted from feet to metres, but the file already gives it in metres: Peterson, for example, is listed at 1,879, which matches its real elevation of about 1,886 m. The conversion has been removed (Field 6, Table 1).
2. **Season.** Every month belongs to exactly one season in the data, so Season adds no information beyond Month and is left out of the model inputs (Field 2).
3. **Related rows.** We added a table of rows and share of good hours per site. The share ranges from 29 percent (Grissom) to 70 percent (JDMT), so rows of one site are related. We now keep whole sites, not only site-days, on one side of every split (Fields 7 and 9).
4. **Baseline.** Measured by cross validation with whole sites in Milestone 2, the baseline is 44.8 percent, not the 50.0 percent estimated from the class shares (Field 4).

Part A. The problem

Field 1. Problem

The users are the energy managers of a large factory that runs partly on its own solar panels. Each day, the day before, they plan which hours to use for energy-hungry but flexible work, such as batch production, cooling, or charging electric forklifts and batteries. For every hour, they decide whether to run this work on solar power or postpone it. Our model supports this decision by predicting whether an hour will be a good solar hour, meaning the panels produce at least 28 percent of their rated power (13.8 of 50 watts on our test panels). For a factory with a 2-megawatt installation, this would mean at least about 550 kilowatts. Our data comes from small test panels, so we use their prediction as a signal for the whole installation: when a test panel has a good hour, the factory's panels, under the same sun and weather, have one too.

The model can be wrong in two ways. If it says "good hour" but the hour is weak, the factory starts work that the panels can't power. It then has to buy electricity from the grid at short notice, often at high prices, or drain its batteries, and production may be delayed. If it says "weak hour" but the hour is good, the factory postpones work it didn't need to, and free solar energy is wasted. The first mistake is worse because it can disrupt production, while the second one only means a missed saving.

Field 2. Task

The target is a new column, good_hour, which we create from the measured power PolyPwr. It is 1 (good solar hour) if the panel produced at least 13.8 watts, and 0 (weak hour) otherwise. Without data on the factory's actual energy needs, we use the median output of 13.8 watts as the

cutoff, so that half of the measurements fall into each class. Our horizontal test panels rarely come close to their rating: the highest reading is 34.3 watts, and only 3 percent of readings reach 25 watts, half the rating. A stricter cutoff would therefore make good hours very rare, so the median marks the better half of the solar hours that these panels actually achieve. The inputs are the place (latitude, longitude, altitude), the time (month and hour) and the weather (humidity, ambient temperature, wind speed, visibility, pressure, cloud ceiling). Season is not used, because it follows completely from Month. The task is supervised, because every row comes with the real answer, taken from the measured power. It is a classification with two classes, because the answer is yes or no rather than a number.

We exclude some columns. PolyPwr is not an input, because the target is made from it, and using it would give the model the answer directly. Location is replaced by latitude, longitude and altitude, which carry the same information. Date and Time are covered by month and hour. YRMODAHRMI is damaged: its day, hour and minute were lost, so it only repeats year and month.

Field 3. Known when

The factory plans its work the day before, so every input must be available at that moment. Latitude, longitude and altitude are fixed for each site. Month, hour and season are known for any hour being planned. The weather inputs (humidity, ambient temperature, wind speed, visibility, pressure and cloud ceiling) were measured at the same time as the power in our data. In real use, the real weather of tomorrow is not known yet, so the factory would take a weather forecast for each hour as an input instead. Forecasts are less accurate than measurements, so the model will likely perform worse in real use than in our tests. This is noted as a risk in Field 9. The measured power PolyPwr is only known after the hour has passed, and the target is made from it, so it is never used as an input. No other column describes what happens after the decision.

Field 4. Success

We measure the model by its accuracy, the share of measurements it classifies correctly, on data it has not seen during training. The baseline is a model that ignores all inputs and always gives the same answer. Because we set the cutoff at the median, the two classes are equal in size. After removing 52 faulty rows (Field 7), there are 10,506 good and 10,487 weak measurements, so this baseline is right for 50.0 percent of them, whichever answer it gives. Measured by cross validation on whole sites in Milestone 2, the baseline turns out lower, 44.8 percent, because hiding some sites tilts the rest toward the other class. Our model must clearly beat it. Because a wrong "good hour" is the worse mistake (Field 1), we add a second measure: the share of weak hours that the model wrongly calls good. This must stay as low as possible, even if that means the model sometimes misses a good hour.

Part B. The data

Field 5. Origin

The data was taken from Mendeley Data, precisely from the dataset "Northern Hemisphere Horizontal Photovoltaic Power Output Data for 12 Sites" (version 5, published 31 July 2019), released under the CC BY 4.0 licence. It was produced by researchers at the Air Force Institute

of Technology together with the University of Dayton, and follows the paper by Pasion et al., "Machine Learning Modeling of Horizontal Photovoltaics Using Weather and Location Data", Energies 13(10), 2570, 2020. The main reason for collecting it was to see whether the power produced by solar panels can be predicted from weather, time and location alone, without costly sunlight measurements.

A small 50-watt panel was connected to a small computer (a Raspberry Pi) at 12 different sites, mainly military bases. It recorded the power, temperature and humidity every 15 minutes. Other data, like wind speed, visibility, pressure and cloud ceiling, came from the nearest airport weather station, up to five miles away. Only the measurements between 10:00 and 15:45 were kept.

Because of this origin, a model could learn some patterns that won't hold in real use. The temperature was measured inside a box in direct sunlight, so it may read higher than the weather forecast would, and partly measure the sunshine itself. Latitude, longitude and altitude have only 12 combinations, so they act like site names, and the model might learn which site tends to be sunny instead of how weather affects power, which does not help at a new location like our factory. Also, some sites were only measured in part of the year, for example Offutt only from April to September, so the model might mix up a sunny site with a sunny season. We check these patterns in Fields 7 and 8.

Field 6. Content

One row is one measurement of one solar panel at one site, taken at 15-minute time stamps between 10:00 and 15:45, with about six measurements per site and day. The data has 21,045 rows and 17 columns, and we add one more, our target good_hour. The file gives wind speed, visibility and cloud ceiling in US units, which we converted to metric. Altitude is already in metres. Table 1 lists all columns with their type, unit and meaning. The target is 1 if the panel produced at least 13.8 watts, the median of PolyPwr, and 0 otherwise. This gives 10,523 good measurements (50.0 percent) and 10,522 weak ones (50.0 percent).

Table 1. Columns

Column	Type	Unit	Meaning
Location	text	–	name of the site
Date	integer	YYYYMMDD	date of the measurement
Time	integer	HHMM	time of the measurement, at 15-minute steps
Latitude	decimal	degrees	north–south position of the site
Longitude	decimal	degrees	east–west position of the site
Altitude	decimal	m (as recorded)	height of the site above sea level
YRMODAHRMI	decimal	–	date and time in one number, damaged (only year and month left)
Month	integer	1–12	month of the measurement
Hour	integer	10–15	hour of the measurement
Season	text	–	Winter, Spring, Summer or Fall
Humidity	decimal	percent	humidity, measured at the panel
AmbientTemp	decimal	°C	temperature, measured at the panel

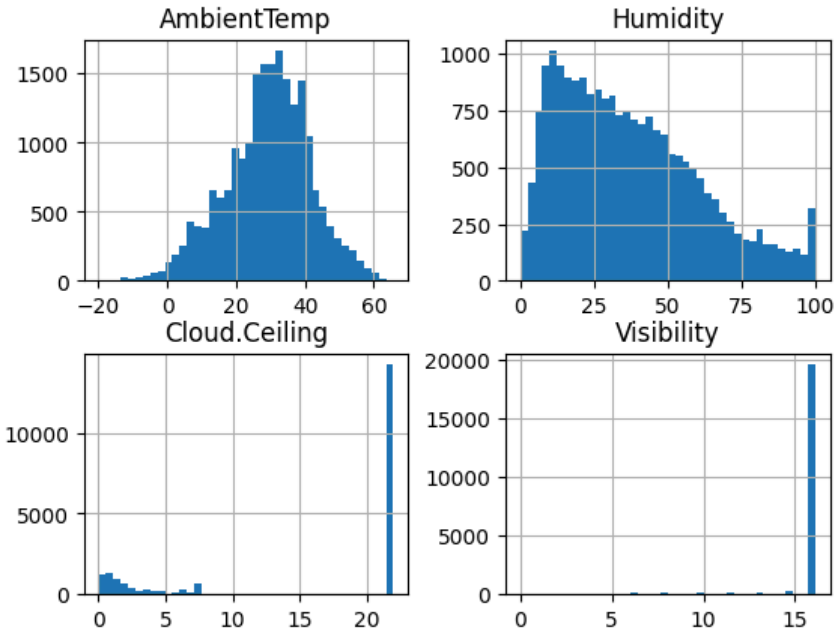
Column	Type	Unit	Meaning
PolyPwr	decimal	watts	power output of the panel, used to make the target
Wind.Speed	decimal	km/h (converted from mph)	wind speed, from the airport station
Visibility	decimal	km (converted from miles)	how far you can see, from the airport station, at most 16.1
Pressure	decimal	millibar	air pressure, from the airport station
Cloud.Ceiling	decimal	km (converted from hundreds of feet)	height of the lowest cloud layer, 22.0 means no clouds
good_hour	integer	0 or 1	target, 1 if PolyPwr is at least 13.8 watts

Field 7. Quality

The data has no missing values and no exact copies, also when comparing only the inputs. Some values, however, are codes or likely errors. At least half of the rows have a cloud ceiling of 22 km. This is not a real height: the paper defines it as the code for “no clouds”, and in our data the highest real cloud is at 9.1 km, with no values between 9.1 and 22 km. At least three quarters of the rows have a visibility of 16.1 km, because the weather stations stop measuring at 10 miles (16.1 km), so visibility carries little information. Humidity is exactly 0 in 52 rows, which is not realistic, so we remove these rows as sensor errors. This leaves 20,993 rows, 10,506 good and 10,487 weak, so the baseline stays at 50.0 percent.

The figure below shows the distributions of four important inputs. Cloud ceiling and visibility are almost entirely at their highest values. Temperature reaches 65.7 °C, far above any real air temperature, which supports our suspicion from Field 5 that the sensor box heats up in the sun. Humidity has a small peak at 100 percent, which may also be a sensor issue. The inputs have very different scales, from 0 to 16.1 km for visibility and 0 to 22 km for cloud ceiling to −20 to 66 °C for temperature.

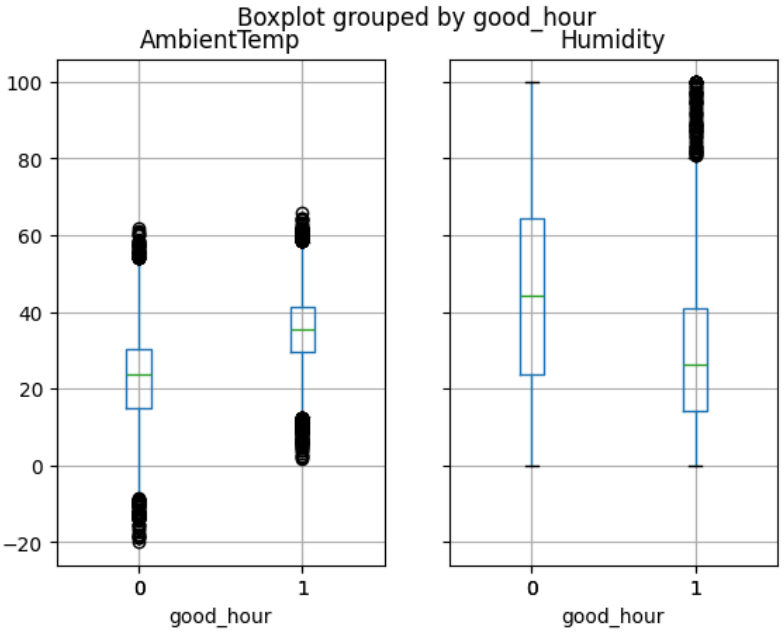
Measurements from the same site on the same day share the same weather and are very similar. There are 3,656 such site-days. Rows of the same site are related too: the share of good hours ranges from 29 percent at Grissom to 70 percent at JDMT. We therefore keep all rows of one site together when we split the data, which also keeps every site-day together. Otherwise the model would be tested on rows almost identical to the ones it was trained on, and the test would be too easy.



Field 8. Signal

We compared good and weak hours on four inputs. Good hours are much warmer, 35.7 °C on average against 22.8 °C, and much drier, with 28.9 percent humidity against 45.6 percent. They also have no clouds in 85 percent of cases, against 50 percent of weak hours. The hour barely differs, 12.8 against 12.5, because all measurements are taken between 10:00 and 15:45, when the sun is already high.

The figure below shows that good hours are clearly warmer and drier, but the two groups overlap a lot, so no single input decides the answer and a model has to combine several of them. The data therefore contains a clear signal that a model can learn. Part of the temperature signal, however, may come from the sensor box heating up in the sun (Field 5), so the model may rely on it more than a weather forecast would allow.



Part C. The risks

Field 9. Risks

Bias from the origin data. The temperature was measured in a box in direct sunlight and reaches 65.7 °C, so it may partly measure sunshine rather than air temperature. A weather forecast would not give such values, so the model may rely on temperature more than it can in real use. Later in the project we will train the model once with and once without temperature and compare the results. Also, latitude, longitude and altitude only identify 12 sites, so the model might learn which site is sunny instead of how weather affects power. Our factory is not one of these sites, so we will test the model on sites it has not seen during training.

Old and limited data. The data comes from 2017 and 2018, from 12 sites in the northern hemisphere, only between 10:00 and 15:45, and from small 50-watt horizontal panels. Some sites, like Offutt, were only measured in summer. The results are claimed only for similar climates, horizontal panels and midday hours, and we assume that the factory's panels react to sun and weather like the test panels.

Missing information. The data has no direct measurement of sunlight, because the original study set out to predict power without costly sunlight sensors. Our inputs only describe the sky indirectly. Cloud ceiling is recorded only when clouds cover most of the sky, and it is measured at an airport up to five miles away. Haze, smoke and dirty or snowy panels are not recorded at all. In these cases, the inputs look sunny while the panel produces little, which leads to our worse mistake, a “good hour” that turns out weak. Our own data shows how common this is: half of the weak hours were recorded with no clouds. We will therefore watch the share of weak hours called good (Field 4) closely, and accept that some of these errors cannot be removed with our inputs.

Leakage and timing. The measured power PolyPwr is only known after the hour, and our target is made from it, so it is never used as an input. The weather inputs are measured values, but in real use the factory would only have a forecast, so the model will perform worse in practice than in our tests. The test results will be presented as a best case and this gap will be openly stated.

Related rows. Measurements from the same site and day share the same weather and are almost identical, and rows of the same site share its location and climate. When splitting into training and test data, we keep all rows of one site together, which also keeps every site-day together, so the test is not too easy.

Our choice of cutoff. We set a good hour at the median of 13.8 watts because we have no data on the factory's real energy needs. A real factory would set its own cutoff, and the model would have to be checked again with it.